

Reverse-Engineering Fixed-Return Assumptions

Taking distributions from an investment portfolio amplifies the impacts of portfolio volatility, making retirement income planning particularly tricky when systematic portfolio distributions tend to be the primary income source for retirees. In financial planning, it is common to use software based on Monte Carlo simulations to show a distribution of potential outcomes for a financial plan by calculating how the plan performs with different sequences of random market returns.

This contrasts with developing a financial plan using a fixed-return assumption in a spreadsheet. Implied investment returns are usually not shown with Monte Carlo simulation output in financial planning software, but they exist underneath the hood. The mistake that is frequently made with a spreadsheet, I worry, is that the choice of return assumptions will reflect a “best guess” about future market returns. This is one of the matters I said we would come back to when describing how to assemble a portfolio return assumption in Exhibit 2.1. This best guess would correspond with only a 50 percent chance for success. Financial planning software generally focuses on having a financial plan work with odds greater than a coin flip. The “best guess” return assumption must be reduced to have greater than a 50 percent confidence that the financial plan will work.

Which implied portfolio return supports a 90 percent chance for success? We have to reverse engineer Monte Carlo simulations to find out. The implied return will be lower than the average returns inputted in the simulation, and I find support for appropriate portfolio return assumptions for the postretirement period being more conservative than for the preretirement period.

Consider three scenarios:

1. An individual investing a lump-sum amount for thirty years,
2. An individual saving a fixed percentage of a constant inflation-adjusted salary at the end of each year over a thirty-year accumulation period, and